```
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

```
df = pd.read_csv('/content/Country-data.csv')
df
```

	country	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp
0	Afghanistan	90.2	10.0	7.58	44.9	1610	9.44	56.2	5.82	553
1	Albania	16.6	28.0	6.55	48.6	9930	4.49	76.3	1.65	4090
2	Algeria	27.3	38.4	4.17	31.4	12900	16.10	76.5	2.89	4460
3	Angola	119.0	62.3	2.85	42.9	5900	22.40	60.1	6.16	3530
4	Antigua and Barbuda	10.3	45.5	6.03	58.9	19100	1.44	76.8	2.13	12200
...	...	...	...	...	...	...	...	...	...	...
162	Vanuatu	29.2	46.6	5.25	52.7	2950	2.62	63.0	3.50	2970
163	Venezuela	17.1	28.5	4.91	17.6	16500	45.90	75.4	2.47	13500
164	Vietnam	23.3	72.0	6.84	80.2	4490	12.10	73.1	1.95	1310
165	Yemen	56.3	30.0	5.18	34.4	4480	23.60	67.5	4.67	1310
166	Zambia	83.1	37.0	5.89	30.9	3280	14.00	52.0	5.40	1460

Next steps: Generate code with df | New interactive sheet

```
print("First 5 records:")
print(df.head())
```

```
First 5 records:
              country  child_mort  exports  ...  life_expec  total_fer   gdpp
0         Afghanistan        90.2     10.0  ...        56.2       5.82    553
1             Albania        16.6     28.0  ...        76.3       1.65   4090
2             Algeria        27.3     38.4  ...        76.5       2.89   4460
3              Angola       119.0     62.3  ...        60.1       6.16   3530
4  Antigua and Barbuda        10.3     45.5  ...        76.8       2.13  12200

[5 rows x 10 columns]
```

```
x = df.drop('country', axis=1)
x
```

	child_mort	exports	health	imports	income	inflation	life_expec	total_fer	gdpp
0	90.2	10.0	7.58	44.9	1610	9.44	56.2	5.82	553
1	16.6	28.0	6.55	48.6	9930	4.49	76.3	1.65	4090
2	27.3	38.4	4.17	31.4	12900	16.10	76.5	2.89	4460
3	119.0	62.3	2.85	42.9	5900	22.40	60.1	6.16	3530
4	10.3	45.5	6.03	58.9	19100	1.44	76.8	2.13	12200
...	...	...	...	...	...	...	...	...	...
162	29.2	46.6	5.25	52.7	2950	2.62	63.0	3.50	2970
163	17.1	28.5	4.91	17.6	16500	45.90	75.4	2.47	13500
164	23.3	72.0	6.84	80.2	4490	12.10	73.1	1.95	1310
165	56.3	30.0	5.18	34.4	4480	23.60	67.5	4.67	1310
166	83.1	37.0	5.89	30.9	3280	14.00	52.0	5.40	1460

167 rows × 9 columns

Next steps: Generate code with x | New interactive sheet

```
scaler = StandardScaler()
x_scaled = scaler.fit_transform(x)
print(x_scaled[:5])
```

```
[[ 1.29153238 -1.13827979  0.27908825 -0.08245496 -0.8082454   0.15733622
  -1.61909203  1.90288227 -0.67917961]
 [-0.5389489  -0.47965843 -0.09701618  0.07083669 -0.3753689  -0.31234747
   0.64786643 -0.85997281 -0.48562324]
 [-0.27283273 -0.09912164 -0.96607302 -0.64176233 -0.22084447  0.78927429
   0.67042323 -0.0384044  -0.46537561]
 [ 2.00780766  0.77538117 -1.44807093 -0.16531531 -0.58504345  1.38705353
  -1.17923442  2.12815103 -0.51626829]
 [-0.69563412  0.1606679  -0.28689415  0.4975675   0.10173177 -0.60174853
   0.70425843 -0.54194633 -0.04181713]]
```

```
x_scaled = pd.DataFrame(
    scaler.fit_transform(x),
    columns=x.columns
)
print(x_scaled.head())
```

```
   child_mort   exports    health  ...  life_expec  total_fer      gdpp
0    1.291532 -1.138280  0.279088  ...   -1.619092   1.902882 -0.679180
1   -0.538949 -0.479658 -0.097016  ...    0.647866  -0.859973 -0.485623
2   -0.272833 -0.099122 -0.966073  ...    0.670423  -0.038404 -0.465376
3    2.007808  0.775381 -1.448071  ...   -1.179234   2.128151 -0.516268
4   -0.695634  0.160668 -0.286894  ...    0.704258  -0.541946 -0.041817

[5 rows x 9 columns]
```

```
kmeans = KMeans(n_clusters = 3, random_state=42, n_init=10)
```

```
kmeans.fit(x_scaled)
```

```
KMeans
KMeans(n_clusters=3, n_init=10, random_state=42)
```

```
df['cluster'] = kmeans.labels_
print("\nCountries with Cluster:")
print(df[['country','cluster']].head(15))
```

```
Countries with Cluster:
                country  cluster
0           Afghanistan        1
1               Albania        2
2               Algeria        2
3                Angola        1
4   Antigua and Barbuda        2
5             Argentina        2
6               Armenia        2
7             Australia        0
8               Austria        0
9            Azerbaijan        2
10              Bahamas        2
11              Bahrain        0
12           Bangladesh        2
13             Barbados        2
14              Belarus        2
```

```
print("\nNumber of Countries in each Cluster:")
print(df['cluster'].value_counts().sort_index())
```

```
Number of Countries in each Cluster:
cluster
0    36
1    47
2    84
Name: count, dtype: int64
```

```
new_sample = pd.DataFrame(
    [[10,40,7,35,30000,5,75,2,15000]],
    columns=x.columns
    )
new_sample_scaled = pd.DataFrame(
    scaler.transform(new_sample),
    columns=x.columns
)
new_cluster = kmeans.predict(new_sample_scaled)
```

```
print("\nNew Country Sample:")
print(new_sample)
print("\nnew country belongs to cluster:")
new_cluster[0]
```

```
New Country Sample:
   child_mort  exports  health  ...  life_expec  total_fer   gdpp
0          10       40       7  ...          75          2  15000

[1 rows x 9 columns]

new country belongs to cluster:
np.int32(2)
```